

Gour Gupal Talukder Shawon

Shahjalal University of Science and Technology, Sylhet, Bangladesh

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Research Interests

Multi-agent LLM systems, AI evaluation and benchmark design, federated learning and privacy-preserving machine learning, large language model alignment, natural language processing, software engineering automation, and distributed systems.

Education

Shahjalal University of Science and Technology

B.Sc. in Software Engineering

Cumulative GPA: 3.45 / 4.00 (through 7th semester)

Sylhet, Bangladesh

Mar 2022 - May 2026

Research Experience

HiveMind Research Group

Research Contributor

Remote

Jan 2026 - Present

Paper: *AgentCollabBench: Diagnosing When Good Agents Make Bad Collaborators* Under Review (NeurIPS 2026)

Resources: [arXiv Preprint](#) | [GitHub](#) | [HuggingFace Dataset](#)

- Contributed to *AgentCollabBench*, a benchmark for evaluating reliability and communication failures in multi-agent LLM systems across Software Engineering, DevOps, and Data Engineering tasks.
- Assisted in early-stage metric and dataset planning for controlled multi-agent evaluation scenarios, including experiments related to inter-agent influence and echo-chamber behavior; created structured JSON benchmark samples and task configurations.
- Reviewed benchmark task structures, communication topologies, and annotation workflows to help improve dataset consistency and evaluation quality across collaborative agent pipelines.
- Participated in collaborative benchmark development and research review processes, contributing to dataset validation, experimental refinement, and technical coordination within a distributed AI research team.

Undergraduate Thesis

Commit Message Generation using LLMs with Issue-Based Context

May 2025 - Nov 2025

Supervisor: Mahfuzur Rahman Emon, Lecturer, IICT, SUST

- Conducted a comparative study of GPT-4 Turbo, Claude 3.5, and Gemini 1.5 Pro on a curated dataset of 309 real-world issue-linked commits from high-authority repositories (freeCodeCamp, OWASP).
- Built a dual evaluation framework combining NLP metrics (BLEU, BERTScore) with double-blind assessment by 30 expert developers, supported by a full-stack Human Evaluation Platform (Next.js, Node.js, MongoDB).
- Achieved 42–53% performance gain in commit message quality over human-written baselines through optimized prompt engineering.

Advanced Machine Intelligence Research Lab, AIUB

Intern Research Assistant

Remote

Feb 2026 - Present

- Contributing to early-stage research on federated unlearning for LoRA-tuned LLMs, focusing on privacy-aware machine learning and GDPR “Right to be Forgotten” compliance in decentralized environments.

- Assisting with literature review, dataset preparation, and preliminary experimental analysis for research involving non-IID federated learning, statistical heterogeneity metrics, and parameter-efficient fine-tuning workflows.
- Participating in an ongoing CVPR-oriented research initiative on spectral restoration of snow-degraded images using diffusion-based models, including proposal development, dataset exploration, and baseline analysis.
- Collaborating within a small research team on experiment planning, technical discussions, and iterative research documentation under faculty supervision.

Industry Experience

ShellBeeHaken Ltd.

Intern Software Engineer (Onsite)

Dhaka, Bangladesh

Dec 2025 - May 2026

- Contributed to production UI components and navigation flows using React Native (Expo) and Next.js, integrating REST APIs and managing application state across screens.
- Assisted in building event-driven backend services with AWS Lambda; gained hands-on exposure to distributed systems concepts including data replication and load balancing during architecture discussions.
- Participated in the full API development cycle from writing data contracts to integration testing, while working in an Agile team with sprint planning and code reviews.

Technical Skills

Languages	JavaScript (ES6+), Java, C++, Python, TypeScript
Frontend	Next.js, React, React Native, Flutter
Backend	Node.js, Express.js, RESTful APIs, GraphQL, Postman
Databases	MongoDB, Firebase, MySQL
Cloud & DevOps	AWS, Docker, Git, Linux, Serverless (AWS Lambda),

Projects

Personality Detection Using Machine Learning

[GitHub](#) | [Live](#)

- Improved model F1 score by 20% via advanced feature engineering and hyperparameter tuning, achieving 91% accuracy in real-time personality prediction.
- Tools: Python, scikit-learn, Flask, Next.js, Pandas, NumPy, joblib.

Local Household Service Provider Platform

[GitHub](#) | [Live](#)

- Designed an optimized data-fetching architecture that reduced retrieval time by 60%, resulting in 20% higher overall application responsiveness and user satisfaction.
- Integrated Google Gemini-powered AI chatbot, improving service query response rate by 60%; authored critical system documents including SRS and project proposal.
- Tools: Next.js, Express.js, MongoDB, Nodemailer, Cloudinary.

IICT Library Management System

[GitHub](#) | [Live](#)

- Built a scalable digital library management system with searchable catalog, lending history, and admin-student workflows; digitized 100% of physical catalog.
- Designed responsive UI in React/TypeScript with secure login and role-based access control.
- Tools: React, Node.js, Express.js, TypeScript, CSS.

Algorithmic Problem Solving

- Solved 670+ [LeetCode](#) problems; maximum streak of **433** days; peak contest rating **1622**; earned 23 badges.
- Completed 250 - 300 algorithmic problems across Codeforces, UVA, and VJudge platforms.

Academic Honors & Certifications

- **2nd Runner-up** - “Bridging Engineering & Computer Science” International Hackathon, organized by ILM-AI, co-hosted by **St. George’s, University of London**. Built a browser-based projectile motion simulator with optional air resistance (two-member team). [GitHub](#) | [Live](#)
- **Certifications:** AWS Cloud Practitioner · AWS Lambda · AWS DynamoDB · AWS API Gateway · Claude Code in Action (Anthropic) · Generative AI (Udacity & AWS) · ML Crash Course (Google).